Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
MERN Stack

M2:Practical – IX

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10.Full Stack Execution and Validation:

a. Integrate React frontend with Node and Express backend.
b. Perform insert, update, and delete operations through UI.
c. Display database records dynamically in React.
d. Implement frontend form validation.
e. Test all APIs using Postman.
f. Execute and verify the complete working of the frontend, backend, and database together.

Code:

server.js:

```
const express = require("express");
const mongoose = require("mongoose");
const cors = require("cors");
const app = express();
app.use(cors());
app.use(express.json());
// Connect to MongoDB
mongoose.connect("mongodb://127.0.0.1:27017/CollegeDB")
  .then(() => {
    console.log("MongoDB Connected Successfully");
    app.listen(3000, () => {
      console.log("Server running on port 3000");
    });
  })
  .catch((error) => {
    console.log("MongoDB Connection Failed:", error.message);
  });

// Student model with validation
const Student = mongoose.model("Student", {
  name: {
    type: String,
    required: [true, "Student name is required"],
    trim: true
  },
  course: {
    type: String,
    required: [true, "Course is required"],
    trim: true
  },
  marks: {
    type: Number,
    required: [true, "Marks are required"],
    min: [0, "Marks cannot be less than 0"],
    max: [100, "Marks cannot exceed 100"]
  }
});
// Validate student data
function validateStudent(data) {
  const { name, course, marks } = data;
  if (
    typeof name !== "string" ||
    !name.trim() ||
    typeof course !== "string" ||
    !course.trim() ||
    marks === "" ||
    marks === undefined ||
    !Number.isFinite(Number(marks)) ||
    Number(marks) < 0 ||
    Number(marks) > 100
  ) {
    return false;
  }
  return true;

}
// GET: Display all students
app.get("/students", async (req, res) => {
  try {
    const students = await Student.find();
    res.json(students);
  } catch (error) {
    res.status(500).json({ message: "Error fetching students" });
  }
```

```
});
// POST: Add a student
app.post("/students", async (req, res) => {
  if (!validateStudent(req.body)) {
    return res.status(400).json({
      message: "Enter a valid name, course and marks between 0 and 100"
    });
  }
  try {
    const student = await Student.create({
      name: req.body.name.trim(),
      course: req.body.course.trim(),
      marks: Number(req.body.marks)
    });
    res.status(201).json({
      message: "Student added successfully",
      student
    });
  } catch (error) {
    res.status(400).json({ message: "Unable to add student" });
  }
});
// PUT: Update a student
app.put("/students/:id", async (req, res) => {
  if (!mongoose.isValidObjectId(req.params.id)) {
    return res.status(400).json({ message: "Invalid student ID" });

  }
  if (!validateStudent(req.body)) {
    return res.status(400).json({
      message: "Enter a valid name, course and marks between 0 and 100"
    });
  }
  try {
    const student = await Student.findByIdAndUpdate(
      req.params.id,
      {
        name: req.body.name.trim(),
        course: req.body.course.trim(),
        marks: Number(req.body.marks)
      },
      { new: true, runValidators: true }
    );
    if (!student) {
      return res.status(404).json({ message: "Student not found" });
    }
    res.json({
      message: "Student updated successfully",
      student
    });
  } catch (error) {
    res.status(400).json({ message: "Unable to update student" });
  }
});
// DELETE: Delete a student
app.delete("/students/:id", async (req, res) => {
  if (!mongoose.isValidObjectId(req.params.id)) {
    return res.status(400).json({ message: "Invalid student ID" });
  }
  try {
    const student = await Student.findByIdAndDelete(req.params.id);

    if (!student) {
      return res.status(404).json({ message: "Student not found" });
    }
    res.json({ message: "Student deleted successfully" });
  } catch (error) {
    res.status(500).json({ message: "Error deleting student" });
  }
});
```

App.jsx:

```
import { useEffect, useState } from "react";
import axios from "axios";
const API = "http://localhost:3000/students";
function App() {
 const [students, setStudents] = useState([]);
 const [name, setName] = useState("");
 const [course, setCourse] = useState("");
 const [marks, setMarks] = useState("");
 const [editId, setEditId] = useState(null);
 const [error, setError] = useState("");
 // Get student records
 const getStudents = async () => {
  try {
   const response = await axios.get(API);
   setStudents(response.data);

   setError("");
  } catch (err) {
   setError("Unable to load students. Check the backend.");
  }
 };
 useEffect(() => {
  getStudents();
 }, []);
 // Clear form
 const clearForm = () => {
  setName("");
  setCourse("");
  setMarks("");
  setEditId(null);
  setError("");
 };
 // Add or update student
 const saveStudent = async (e) => {
  e.preventDefault();
  setError("");
  // Validation 1: Empty fields
```

```
    if (!name.trim() || !course.trim() || marks === "") {
      setError("Please fill in all fields.");
      return;
    }
    // Validation 2: Marks must be between 0 and 100
    if (
      !Number.isFinite(Number(marks)) ||
      Number(marks) < 0 ||
      Number(marks) > 100
    ) {
      setError("Marks must be between 0 and 100.");
      return;
    }
    const student = {
      name: name.trim(),
      course: course.trim(),
      marks: Number(marks)
    };
    try {
      let response;
      if (editId) {
        response = await axios.put(`${API}/${editId}`,
student);
      } else {
        response = await axios.post(API, student);
      }
      alert(response.data.message);
      clearForm();
      await getStudents();
    } catch (err) {
      setError(
        err.response?.data?.message || "Unable to save
student."
      );
    }
  };
  // Load details for editing
  const editStudent = (student) => {
    setEditId(student._id);
    setName(student.name);
    setCourse(student.course);
    setMarks(String(student.marks));
    setError("");
  };
  // Delete student
  const deleteStudent = async (id) => {
    if (!window.confirm("Do you want to delete this
student?")) {
      return;
    }

    try {
      const response = await axios.delete(`${API}/${id}`);
      alert(response.data.message);
      if (editId === id) {
        clearForm();
      }
      await getStudents();
    } catch (err) {
      setError(
        err.response?.data?.message || "Unable to delete
student."
      );
    }
  };
  return (
    <div>
      <h1>Student Management System</h1>
      <h2>{editId ? "Update Student" : "Add Student"}</h2>
      <form onSubmit={saveStudent}>
        <input
          type="text"
          placeholder="Student Name"
          value={name}
          onChange={(e) => setName(e.target.value)}
        />
        <input
          type="text"
          placeholder="Course"
          value={course}
          onChange={(e) => setCourse(e.target.value)}
        />
        <input
          type="number"
          placeholder="Marks (0-100)"

          min="0"
          max="100"
          step="any"
          value={marks}
          onChange={(e) => setMarks(e.target.value)}
        />
        <button type="submit">
          {editId ? "Update Student" : "Add Student"}
        </button>
        {editId && (
          <button type="button" onClick={clearForm}>
            Cancel
          </button>
        )}
      </form>
      {error && <p style={{ color: "red" }}>{error}</p>}
      <h2>Student List</h2>
      <table border="1" cellPadding="8">
        <thead>
          <tr>
            <th>Name</th>
            <th>Course</th>
            <th>Marks</th>
            <th>Actions</th>
          </tr>
        </thead>
        <tbody>
          {students.length === 0 ? (
            <tr>
              <td colSpan="4">No students available.</td>
            </tr>
          ) : (
            students.map((student) => (
              <tr key={student._id}>

                <td>{student.name}</td>
                <td>{student.course}</td>
                <td>{student.marks}</td>
```

```
            <td>                                                    Delete
              <button                                              </button>
                type="button"                                     </td>
                onClick={() => editStudent(student)}             </tr>
              >                                                   ))
                Edit                                             )}
              </button>                                         </tbody>
              <button                                          </table>
                type="button"                                 </div>
                onClick={() => deleteStudent(student._id)}     );
              >                                               }
export default App;
```

Output:







